📬 Project Title: WhisperChain+: Anonymous Messaging with Role-Based Control and Accountability

Duration: 4 Weeks (1 Month)

🎯 Project Backstory: Kindness Meets Control

Imagine this: You're part of a thriving learning community—students, TAs, faculty—each playing different roles. Within this group, people do incredible things every day: ask brave questions, give quiet support, stay late helping others. But these moments often go unacknowledged.

What if you could anonymously send a note of encouragement or thanks to someone without anyone knowing it was you? What if you could do it **without trusting a central authority**, while still keeping the system safe from abuse?

That's where **WhisperChain+** comes in.

This project challenges you to design a secure, **anonymous compliment delivery platform**—but with a twist: you must build in **real-world cybersecurity mechanisms** like **Role-Based Access Control (RBAC)**, **audit logging**, and **accountability**. The system must be **fair, tamper-resistant, and abuse-aware**—without compromising user anonymity.

It's not just an app. It's a lesson in building **usable, trustworthy, privacy-preserving systems.**

🧩 Core Mission

Build a role-based anonymous messaging platform that:

- Allows users to send **anonymous, encrypted compliments**
- Enforces **strict role-based permissions** for sending, receiving, flagging, and auditing

- Supports **logging and moderation** without breaking anonymity
- Prevents spam, message flooding, and identity abuse
- Maintains privacy while offering **traceability and accountability**

👥 Roles in the System

Role	Capabilities
Sender	Can send one encrypted, anonymous message per round
Recipient	Can receive and decrypt messages sent to them
Moderator	Can view flagged messages and review audit logs
Admin	Manages registration, roles, public keys, and can rotate system policies

Each role must be clearly enforced in your access control logic (e.g., ACL or RBAC).

🗓️ Project Timeline & Milestones

Week 1: Design & Threat Modeling

- Identify all possible actions and map them to roles
- Create a **role-permission matrix**
- Design your **access control implementation**: ACLs, capabilities, or rule-based
- Write a **threat model**:
    - What happens if the admin is compromised?
    - What if tokens are shared?

    - How do you ensure nonrepudiation?
- Submit: design doc + RBAC table + threat model draft

Week 2: Token Issuance + Role Enforcement

- Require users to register with an assigned role
- Issue **anonymous, unlinkable message tokens** (1 per round)
    - E.g., blind signature or pseudonymous ID
- Implement checks so that:
    - Only Senders can submit messages
    - Only Moderators can view flagged content
    - Admin cannot view unflagged messages or sender info

Week 3: Messaging & Secure Delivery

- Each user generates a public/private key pair at signup
- Compliments are encrypted using the recipient's **public key**
- Messages stored server-side with no readable metadata
- Recipients retrieve and **decrypt messages locally**
- All actions (token issued, message sent, message flagged) must be logged:
    - Log time, action type, and role involved
    - Do **not** log sender identity

Week 4: Moderation, Logging, and Final Integration

- Allow recipients to **flag abusive messages**

- Flagged messages go to **moderator queue**
- Moderators can **freeze tokens** to block future abuse from anonymous sources
- Implement alerting rules (e.g., threshold for flagged messages)
- Deliverables:
    - Working system + audit log samples
    - Security report
    - Final threat model
    - Optional: UI demo or walkthrough

✅ Must-Have Features

- One anonymous, encrypted message per sender per round
- End-to-end encryption using RSA or ECC
- RBAC controls: enforce role permissions for all actions
- Append-only, timestamped audit log
- Flagging + moderation system
- Separation of duties: admins can't see content, only meta-events

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🔐 Security & Privacy Requirements

- Use cryptographic libraries (Python `cryptography`, Web Crypto API, etc.)
- Ensure message tokens are unlinkable to identities

- Encrypt messages at rest and in transit
- Log only role-level metadata (e.g., “Sender token used”, not sender name)
- (Optional) Sign audit logs with system key to prevent tampering

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🧠 Advanced Extensions

💡 Undergraduates (Optional Bonus, +10%)

- Blind signature token issuance
- Tamper-proof logs with Merkle trees
- Front-end that adapts to roles (moderator tools, recipient inbox, etc.)
- Intrusion detection rule (e.g., too many messages flagged from one token)
- Public-facing, anonymized dashboard

🎓 Graduate Students (Required Advanced Feature)

Choose and implement **at least one** of the above. You must:

- Justify the design and explain its integration
- Reflect on its trade-offs (privacy, usability, complexity)
- Demonstrate it clearly in your deliverables

📦 Final Submission Checklist

1. **GitHub repository** (clean code, README, setup instructions)

2. **Security design report** (3–5 pages, system + rationale)
3. **RBAC matrix + threat model** (as tables or diagrams)
4. **Sample audit log output** with brief analysis
5. (Optional) UI walkthrough or demo recording
6. **Graduate feature doc** (for graduate students only)

📘 Learning Outcomes

- Implement secure communication with end-to-end encryption
- Design and enforce role-based access policies (RBAC)
- Build auditing mechanisms that preserve privacy
- Balance usability and traceability in complex systems
- Think like an adversary—and defend against one

This isn't just a project. It's a reflection of how real-world systems must balance **human behavior**, **security policies**, and **ethical design**.

You're building a platform for invisible kindness—but it must be **tough enough** to handle abuse, **smart enough** to protect privacy, and **clear enough** to enforce trust without breaking it.

📬 WhisperChain+ is where usability meets cryptographic justice.

Now go make something awesome.

Good luck!

WhisperChain+ Project Timeline, Milestones, and Grading

Overview

The WhisperChain+ project will be graded based on **two main components**:

Component	Points	Weight
Final Oral Evaluation	180	90%
Weekly Check-ins (Milestones)	20	10%
Total	200	100%

Weekly Milestone Check-ins (20 points total)

- Each weekly check-in is worth **5 points**.
- To earn full points:
    - Your team must show **substantial progress** toward that week's milestone **or**
    - Come prepared with **thoughtful questions** and discussion about your design and challenges.

Weekly check-ins are to **help you stay on track** — and feedback will be given weekly to support your progress.

Project Timeline and Weekly Milestones

Week 1: April 25 – May 1

Milestone 1: System Design and Threat Modeling

Deliverables (Due Thursday, May 1, 2025, by 11:59 PM):

- **System Overview** (~1–2 pages) explaining how your WhisperChain+ platform will work
- **Defined Roles** (Sender, Recipient, Moderator, Admin) with clear descriptions
- **Role-Permission Matrix** showing which roles can perform which actions
- **Initial Threat Model**:
    - Identify at least **3 realistic threats** (e.g., token abuse, admin compromise)
    - Briefly explain defenses against each threat
- **Access Control Plan**: Specify whether you use ACLs, capability-based security, or another model

Goal: A full, written security design to guide implementation.

Week 2: May 2 – May 8

Milestone 2: Tokenization and RBAC Implementation

Deliverables (Due Thursday, May 8, 2025, by 11:59 PM):

- **User Registration System** with role assignments
- **Anonymous Token Generation**:
    - Each Sender receives **one anonymous token per round**, unlinkable to their real identity
- **Role Enforcement**:

    - Implement access control checks (who can send, who can view, who can moderate)

- **Audit Logging Setup**:
    - Start recording token issuance, message sending, and flagging events

Goal: User accounts, roles, token issuance, and access control working.

Week 3: May 9 – May 15

Milestone 3: Secure Messaging and Encryption

Deliverables (Due Thursday, May 15, 2025, by 11:59 PM):

- **Public/Private Key Generation** for each user
- **Message Submission Form** using tokens
- **Message Encryption**:
    - Messages encrypted with recipient's public key
- **Recipient Inbox**:
    - Recipients decrypt their own messages
- **Audit Log Updates**:
    - Record message delivery events with timestamps and role info

Goal: Secure, encrypted message flow between users.

Week 4: May 16 – May 23

Milestone 4: Moderation, Logging, and Final Integration

Deliverables (Due Thursday, May 23, 2025, by 11:59 PM):

- **Flagging System**:
    - Recipients can flag inappropriate messages
- **Moderator Dashboard**:
    - Moderators can review flagged messages and freeze abusive tokens
- **Complete Audit Logging**:
    - Record all major events with timestamps
    - Logs must be tamper-resistant (e.g., append-only, digitally signed if possible)
- **Final Threat Model and Updated Design Report**
- **Final Codebase** (well-documented) and optional **Demo Video**

Goal: A complete, polished WhisperChain+ system ready for oral evaluation.

Weekly Summary Table

Week	Dates	Focus	Deliverables Due
Week 1	Apr 25 – May 1	System Design + Threat Model	May 1, 2025
Week 2	May 2 – May 8	Token System + Role Enforcement	May 8, 2025
Week 3	May 9 – May 15	Secure Messaging + Encryption	May 15, 2025
Week 4	May 16 – May 23	Moderation + Final Integration	May 23, 2025

Final Oral Evaluation (180 points total)

The final oral evaluation will assess your completed WhisperChain+ system.
Here's the grading breakdown:

Category	Points
System Functionality	60
Security Design and Threat Model	40
Role-Based Access Control (RBAC)	30
Encryption and Privacy Enforcement	30
Audit Logs and Accountability	20

Graduate students must also present and explain their **advanced feature** as part of the oral evaluation.

Full credit is based on demonstrating a working, complete, secure system — including anything that was originally missed earlier.

Reminder to Students

- Weekly check-ins **are graded** (5 points each), making up **10% of the project grade**.
- Your final oral evaluation is **the main grading event** (90%).
- Submitting empty or placeholder work at milestones will hurt milestone scores.
- Staying engaged and building consistently week by week will make your final evaluation much easier.

- Falling too far behind will make finishing on time very difficult — use the structure to your advantage!